CS & IT ENGINEERING

Computer Network

MAC Sub-Layer

Lecture No. - 01

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ABOUT ME

Hello, I'm Abhishek

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Topics to be Covered

Topic Multiple Access Protocol

Topic Pure ALOHA

Topic : Connection Type

→ Attachment of communication devices to a link.

→ Two possible types of connections :

1. Point-to-point connection
2. Multi-point (multidrop) connection

Topic : Point-to-point connection

→ Dedicated link between two device
[One sender and one receiver]

Topic : Data flow

-> Communication between two devices can be :

1. Simplex mode
   -> One-way communication

2. Half-Duplex mode
   -> Either side communication at a time

3. Full-Duplex (Duplex) mode
   -> Both side communication is possible at same time

Topic : Multi-point connection

→ Multi-point (multidrop) connection

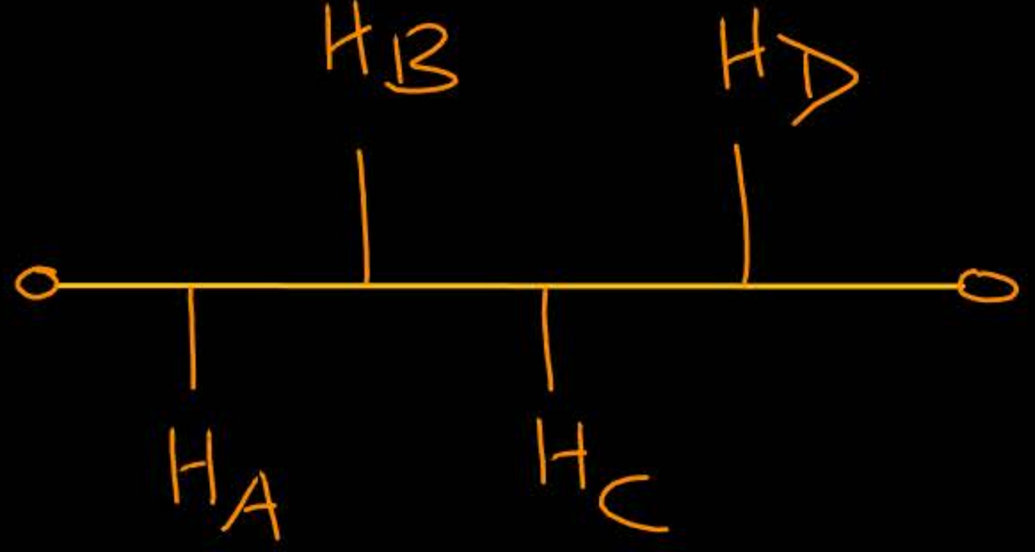
→ More than ~~two~~ one devices share a single link

→ Broadcast medium

[One sender and all are receiver]

→ e.g. Bus topology, Wireless Communication

WiFi



Topic : Network Topology

Star:
Hub, Switch, $\boxed{\text{Router}}$

→ Arrangement of hosts inside a network

→ Different types of topology are :

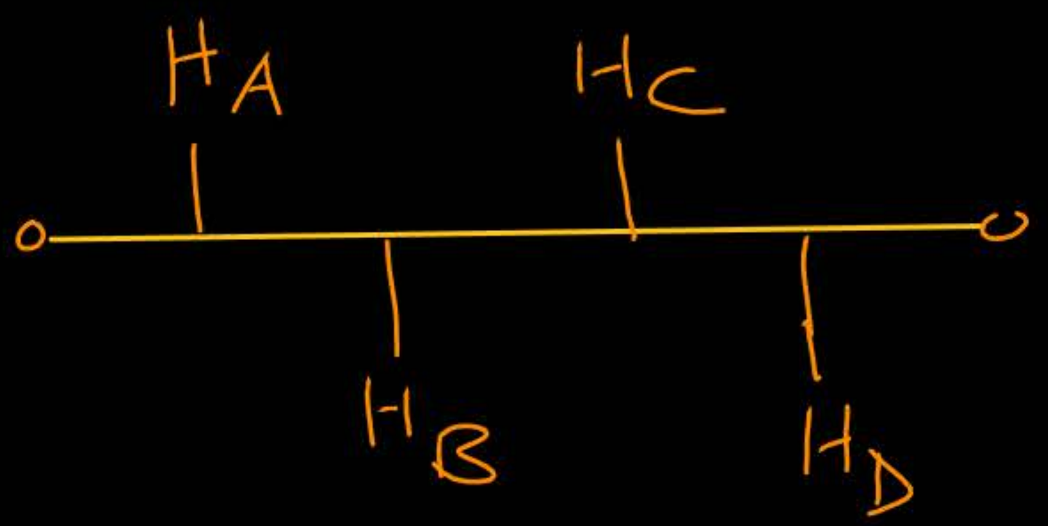
1. Mesh ⟶ P-P link
2. Star ⟶ P-P link
3. Bus ⟶ Multipoint
4. Ring ⟶ P-P link

Topic : Bus Topology

→ Every host connected to centralized backbone media (coaxial cable).

Total number of nodes = n

Total number of links = 1



→ Multipoint [Multidrop]

→ Access Control Method

⇓

Multiple Access

Topic : Bus Topology

Advantages :-

-> Installation cost is very low

[Preferable for long area network]

Disadvantages :-

-> If backbone media fails

then all communications are stop

Topic : Data Link Layer

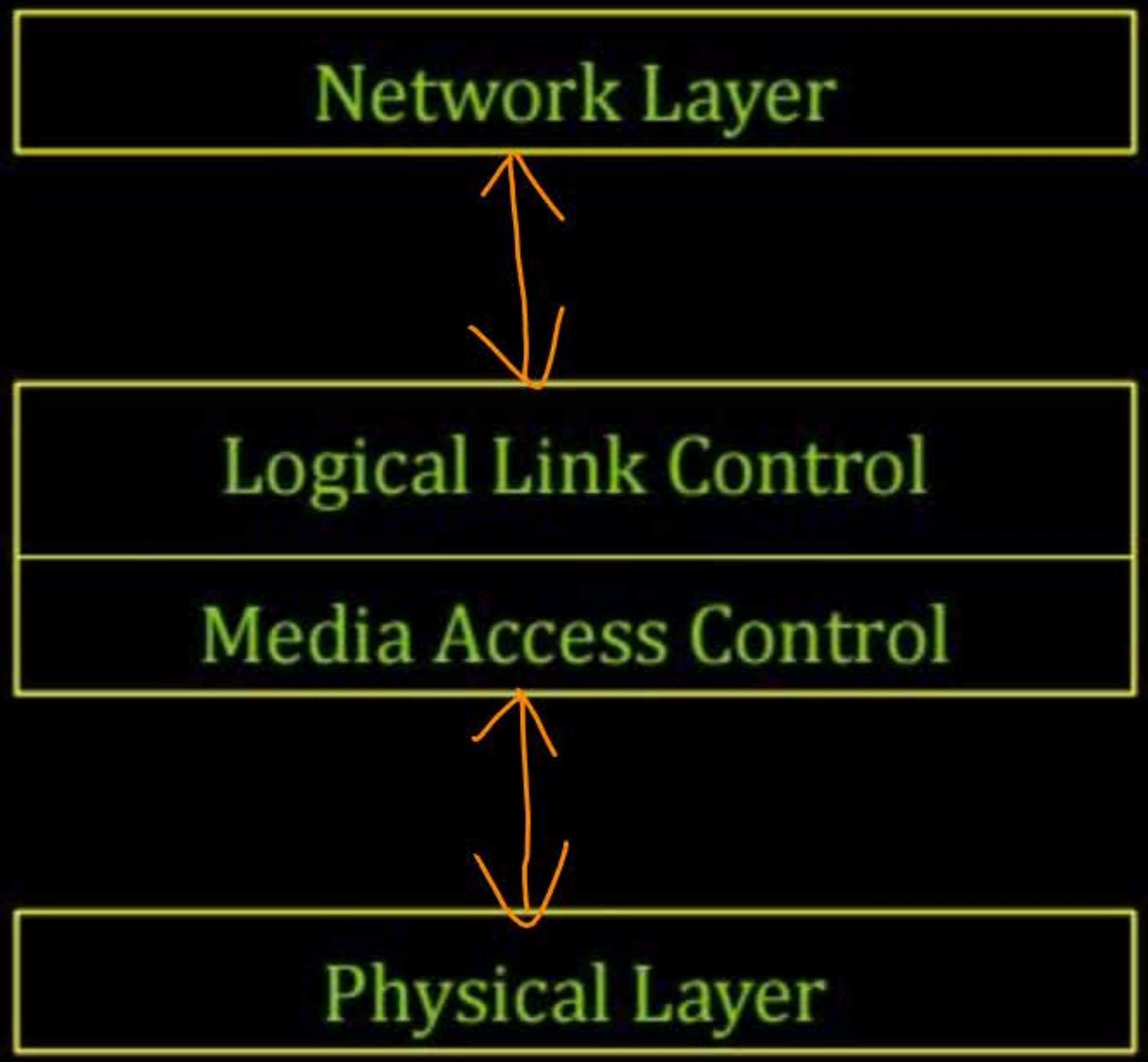
Data link layer services :

1. Framing ⟹ Frame Synchronization (Bit stuffing)
2. Error Control ⟹ 2D parity, Hamming Code
3. Flow Control : Point to Point Communication ⟹
4. Access Control : Multipoint communication

Topic : Data Link Layer

Data link layer is divided into two sub layers :

1. Logical link control (LLC)
2. Media access control (MAC)



Topic : Logical Link Control

-> The upper sub layer

-> Responsible for data link control

-> Flow and Error control

Topic : Media Access Control

-> The lower sub layer

-> Determine "who can access the media"
[in multipoint (multidrop) line configuration]

-> No any use, in point-to-point line configuration

-> Multiple access protocols

Topic : MAC Address

-> Identification of Source and Destination nodes of a packet

-> Physical adddress [Hardware Address]
[NIC : Network Interface Card]

-> 48 bits or 6 bytes

-> Should be unique (worldwide)

Topic : MAC Address

-> Two representation of MAC address :

1. Binary Representation [48-bits]
2. Hexadecimal Representation

e.g. "ac:d1:b8:d5:59:0d"

-> Broadcast MAC address (Special MAC address) :

"ff:ff:ff:ff:ff:ff" [1111..1111]

48-bits one

Topic : Multiple Access Protocol

Interference :

-> Two or more simultanious transmssion by nodes

Collision :

-> if node receives two or more signals at same time

Topic : Multiple Access Protocol

Distributed algorithm :

-> Determine "how node shares channel"

-> Determine "when node can transmit"

-> Communication about channel sharing must use channel itself

-> Fully decentralized : No any special node to coordinate

Topic : Multiple Access Protocol

MAC protocols are divided into three categories :

1. Channel Partitioning : Static Allocation
2. Random Access : No any Allocation
3. Taking Turns : Dynamic Allocation

No collision

Chance of collision

Topic : Channel Partitioning

-> Channelization protocols

-> Divide channel into smaller pieces
[Time slot, Frequncy or Code]

-> Allocate piece to node for exclusive use

Topic : Channel Partitioning

Channel partitioning MAC protocols are :

1. TDMA : Time-Division Multiple Access
2. FDMA : Frequency-Division Multiple Access
3. CDMA : Code-Division Multiple Access

Topic : Random Access

-> Channel is not divided
[unlike channel partitioning]

-> When node has packet to transmit

Topic : Random Access

-> Random Access MAC protocols are :

1. ALOHA

    -> Pure ALOHA and Slotted ALOHA

2. CSMA : Carrier Sense Multiple Access

    -> CSMA/CD [for wired] and CSMA/CA [for wireless]

Topic : Taking Turns

Channel partitioning

-> Static allocation

-> No chance of collision

-> Efficient at high traffic/load [Inefficient at low traffic/load]

Random Access

-> No any allocation

-> Chance of collision

-> Efficient at low traffic/load [Inefficient at high traffic/load]

Topic : Taking Turns

-> Controlled Access Protocol

-> Channnel allocated explicitly
[No chance of "collision"]

-> Dynamic allocation

Topic : Taking Turns

-> Controlled Access MAC protocols are :

1. Reservation
2. Polling
3. Token Passing

2 mins Summary

Topic Multiple Access Protocol

Topic ~~Pure ALOHA~~

THANK - YOU

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